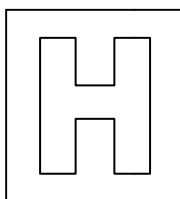


Candidate Name: _____

Class Adm No

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2014 Promotional Examination II Pre-university 2

H1 Biology

8875/02

17 September 2014

2 hours

Additional Materials: Writing papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, Adm No. and class on all the papers you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions on the answer booklet

Section B

Answer **one** question on the writing papers.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets[] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Section B	
Total	

This question paper consists of 16 printed pages including 1 blank page

[Turn over

Section A

Answer **all** questions in this section.

1. Lactate dehydrogenase is an enzyme that is found extensively in body tissues such as the blood cells and muscle cells. Fig. 1.1 shows the structure of human lactate dehydrogenase.

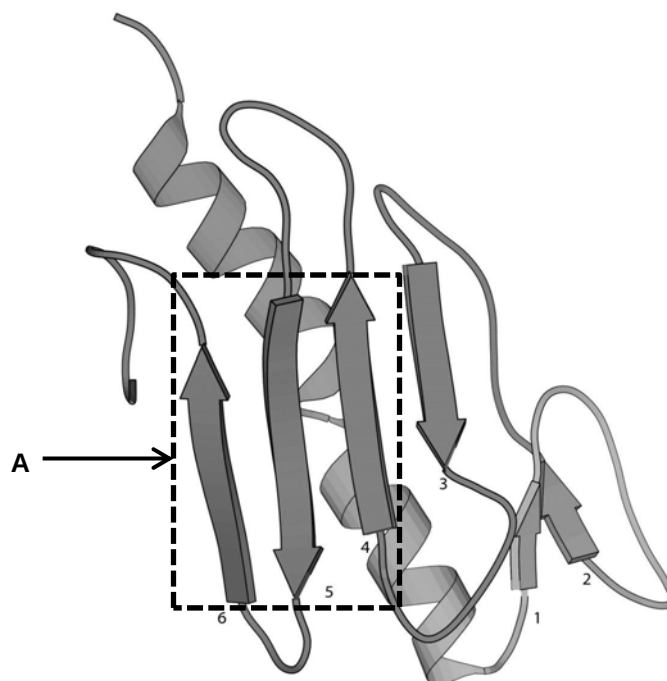


Fig. 1.1

- (a) (i) Identify structure **A**.

.....[1]

- (ii) Describe how structure **A** is formed.

.....

.....[1]

Lactate dehydrogenase works by binding to its substrate pyruvate, which is a product of glycolysis. Lactate dehydrogenase catalyses the conversion of pyruvate into lactate. The toxic effects of heavy metals on organisms stems from their effect on metabolically-important enzymes, such as lactate dehydrogenase. Fig. 1.2 shows the effect of varying concentrations of mercury ions on the rate of reaction catalysed by lactate dehydrogenase isolated from a species of fish.

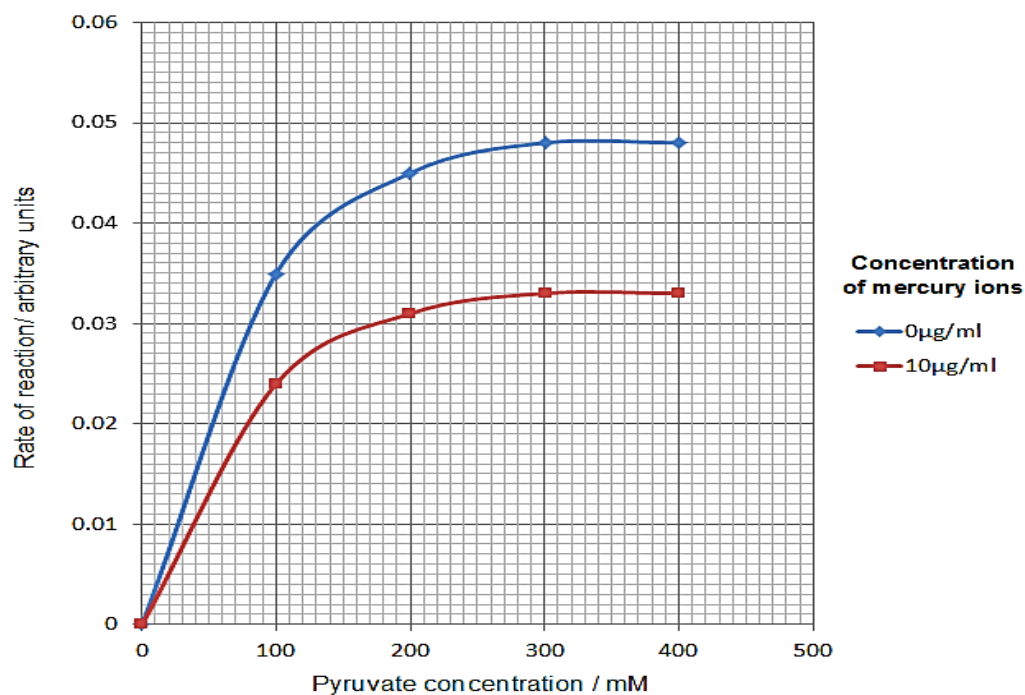


Fig. 1.2

(b) (i) Briefly describe the mode of action of lactate dehydrogenase.

.....

[2]

(ii) With reference to Fig. 1.2, describe and explain the effect of mercury ions on the rate of reaction catalysed by lactate dehydrogenase.

.....

[3]

In nature, collagen is found exclusively in animals, especially in the flesh and connective tissues of mammals. Collagen is a part of the connective tissues in the skin that helps in maintaining skin firmness and suppleness. Collagen is a fibrous protein and has a very distinct structure and function from lactate dehydrogenase.

(c) Contrast between the structure of collagen and lactate dehydrogenase.

.....

.....

.....

.....[2]

[Total: 9]

2. Fig. 2.1 is an electron micrograph of chromatin found in the nuclei of eukaryotes. Most of the chromatin is seen to be in the form of a fiber with a diameter of about 30 nm.



Fig. 2.1

- (a)** Describe how DNA is packed into the 30 nm chromatin fibre.

.....

.....

.....[2]

In the 1930s, Hermann Muller was the first researcher to note that the ends of chromosomes had unique properties. Muller named these ends telomeres (from the Greek words *telo*, meaning "end," and *mere*, meaning "part"), based on their position on chromosomes.

There is a limit to the number of times a normal human cell population will divide. Empirical evidence shows that telomeres get slightly shorter with each round of DNA replication until they shorten to a critical length to trigger cell death.

Fig. 2.2 shows the position of telomeres in a DNA double helix.

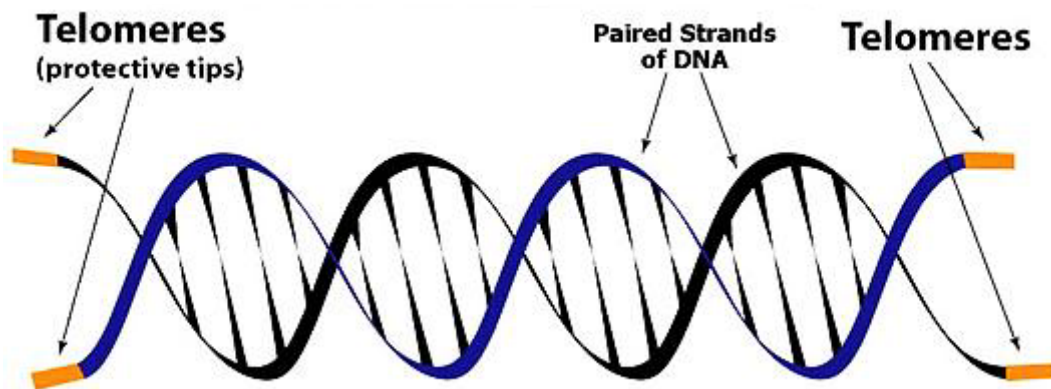


Fig. 2.2

(b) Explain why telomeres shorten with each round of DNA replication.

.....

.....

.....

.....[2]

(c) State one role of telomeres in cells.

.....

.....[1]

Fig. 2.3 and Fig. 2.4 show the process of DNA replication and transcription in eukaryotes

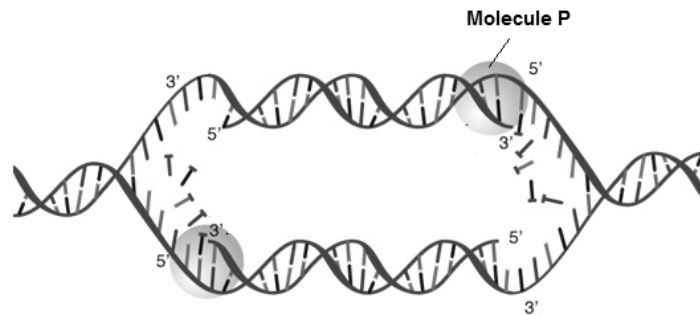


Fig. 2.3

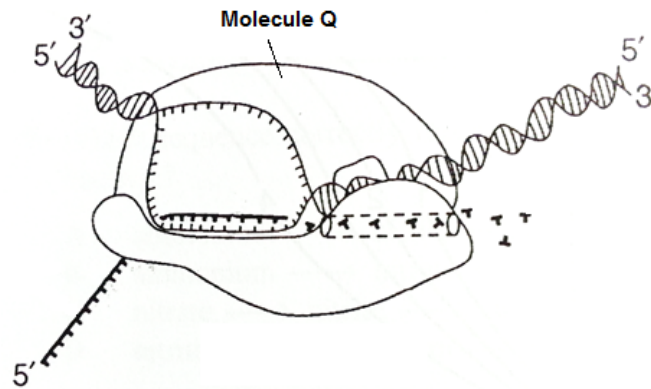


Fig. 2.4

- (d) Explain why RNA primer is required for the synthesis of new strand in DNA replication.

.....
[1]

- (e) With reference to Fig. 2.3 and Fig. 2.4, describe **two** ways in which the functions of molecules **P** and **Q** are similar to each other.

.....

[2]

[Total: 8]

3. The term 'Manx syndrome' is used to describe a range of conditions seen cats with a shortened or absent tail. This syndrome arises as a result of a gene mutation in Manx gene. Manx gene determines the development of the cat's spine. Unfortunately, this mutation is often linked to some severe developmental abnormalities in the spinal cord leading to significant neurological disease.

Fig. 3.1 shows the sequence of bases in a section of Manx gene coding for the first seven amino acids of Manx polypeptide chain.

5' AAAGTGTCTCTGCTGGAACAT 3'

Fig. 3.1

- (a) (i) Using the space below, list the corresponding codons in the mRNA transcribed from the template in Fig. 3.1.

[1]

- (ii) Explain why a single base deletion in the DNA sequence is more likely to produce non-functional proteins than a single base substitution.

.....

[2]

- (iii) Describe how mutation in Manx gene results in the abnormal phenotype in cats with Manx syndrome.

.....

[2]

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The allele S^N codes for normal spine which results in long tail being form and S^A codes for abnormal and short spine. Cats that are homozygous recessive will not have their spines developed and so they will die early in development. Manx cats are tailless and they have a heterozygous genotype. Fig. 3.2 shows a cat with Manx syndrome.



Fig. 3.2

A second gene which controls for the appearance of the coat is located on the X chromosome. Allele D codes for 'dappled' coat (D) while allele d codes for 'plain' coat (d). A male Manx cat with dappled coat was mated on several occasions to the same female and a large number of offspring consisted of the following phenotypes, as summarised in Table 3.1 is observed.

Table 3.1

Tail Type	Gender	Coat colour	% no. of kittens
Long tailed	Male	Dappled	12.5
Long tailed	Female	Dappled	12.5
Manx	Male	Dappled	25
Manx	Female	Dappled	25
Did not survive development			25

- (b)** Using suitable symbols, construct a genetic diagram to explain the results in Table 3.1.

[4]

[Total: 9]

4. Red fox and kit fox are two different species of fox that belong to the family of family Canidae, which are found commonly in North America.

The red fox, *Vulpes vulpes*, resembles a small dog with a long bushy tail, large pointed ears, a pointed muzzle and a red coat. On the other hand, the kit fox, *Vulpes macrotis*, is the smallest species of the family. It has large ears that help the fox lower its body temperature and give it exceptional hearing. Red fox can be found in the farmlands and forests of North America as compared to the kit fox, which lives on the plains and in deserts.

Through a series of studies, the red fox and kit fox are found to be evolved from a common ancestor.

- (a) Explain how natural selection could have caused the evolution of red fox and kit fox in North America.

.....

.....

.....

.....

.....

.....[3]

Our understanding of the evolutionary relationship of species has been greatly improved through the use of molecular techniques. Many evolutionary studies use mitochondrial DNA rather than nuclear DNA.

- (b)** Explain why mitochondrial DNA is used to determine phylogeny instead of using nuclear DNA.

.....
.....
.....[2]

- (c)** Explain how you could use molecular homology to show that red fox and kit fox evolved from a common ancestor.

.....
.....
.....[2]

[Total: 7]

5. Dwarfism is genetic condition characterised by restricted growth and short stature. The average adult height among people with dwarfism is 122 cm. A significant proportion of dwarfism-related disorders are due to growth hormone deficiency. Growth hormone deficiency is treated with injections of a synthetic version of human growth hormone (hGH).

Fig 5.1 shows the pCMV6-XL5 plasmid. It is a plasmid with a multiple cloning site (MCS) that lies downstream of the CMV promoter. This plasmid can be inserted into both eukaryotic and prokaryotic host cells. The arrows denote the direction in which the genes are transcribed.

An artificially-synthesised hGH gene with flanking *HindIII* restriction site sequences was created. The restriction sites for the restriction enzymes *HindIII* and *SacI* are shown in Fig. 5.2.

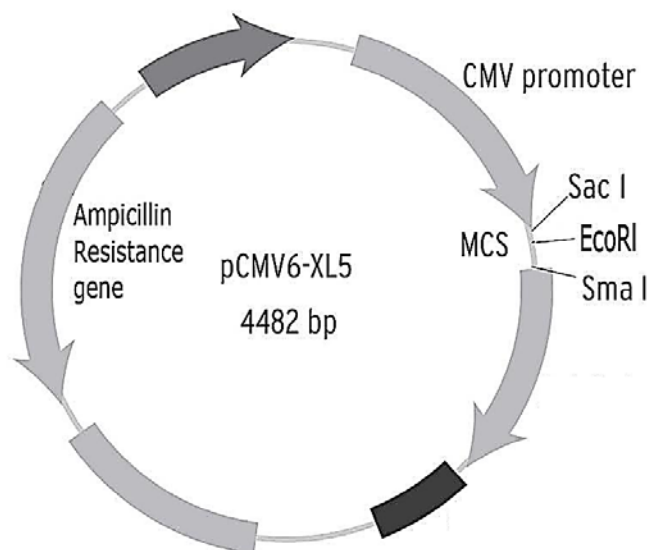


Fig. 5.1

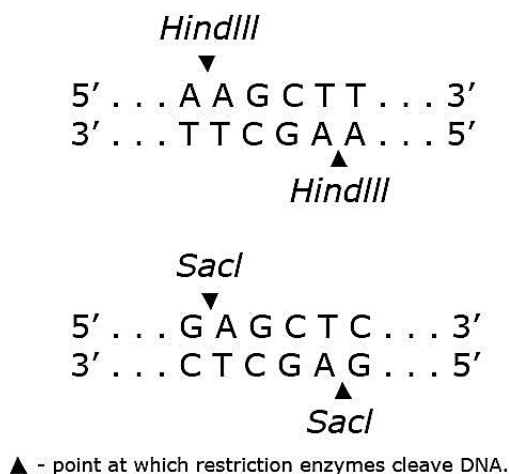


Fig. 5.2

- (a)** Using information in Fig. 5.1 and 5.2, explain how the hGH gene can be inserted into pCMV6-XL5.

.....

.....

.....

.....

.....[3]

- (b)** Outline the steps required to select bacterial colonies containing the human growth hormone gene after transformation has taken place.

.....

.....

.....

.....

.....[3]

- (c)** State one problem of cloning human genes in bacteria.

.....

.....[1]

[Total: 7]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc. as indicated in the question.

- 6.
- (a) Distinguish between the behaviours of homologous chromosomes in mitosis and the first meiotic division. [8]
 - (b) With reference to specific genes, discuss how mutations can lead to cancer. [6]
 - (c) Relate the structure of cell surface membrane to its properties. [6]
- [Total: 20]
- 7.
- (a) Describe the main features of oxidative phosphorylation. [7]
 - (b) Describe the main stages of the Calvin cycle. [6]
 - (c) Discuss the social implications of genetically modified organisms. [7]
- [Total: 20]

End of Paper